Duty Cycle (Duration of Use)

Avoid damage to the Welder by not welding for more than the prescribed duty cycle time. The Duty Cycle defines the number of minutes, within a 10 minute period, during which a given welder can produce a particular welding current without overheating.

For example, a welder with a 40% duty cycle at 125A welding current must be allowed to rest for at least 6 minutes after every 4 minutes of continuous welding.

Failure to carefully observe duty cycle limitations can easily over-stress a welder's power generation system contributing to premature welder failure.

This Welder has an internal thermal protection system to help prevent this sort of over-stress. When the Welder overheats, it automatically shuts down and a warning screen appears in the LCD Display window. The Welder automatically returns to service after cooling off. Should this occur, rest the Tig Torch or Electrode Holder on an electrically non-conductive, heat-proof surface, such as a concrete slab, well clear of the ground clamp.

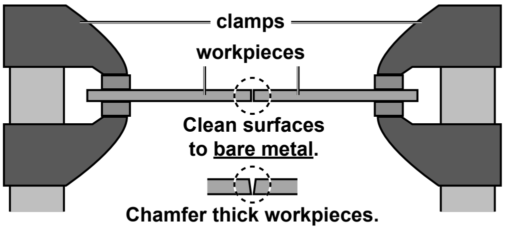
Allow the Welder to cool with the Power Switch on, so that the internal Fan will help cool the Welder.

When normal operation resumes, use shorter welding periods and longer rest periods to prevent needless wear.

TIG Rated Duty Cycles	
240VAC **30% Use at 175A** **For 10 Continuous Minutes**	**120VAC** **40% Use at 125A** **For 10 Continuous Minutes**
3 Minutes Welding / **7 Minutes Resting**	**4 Minutes Welding** / **6 Minutes Resting**
100% Continuous Use at 105A	**100% Continuous Use at 90A**

Stick Rated Duty Cycles	
240VAC **25% Use at 175A** **For 10 Continuous Minutes**	**120VAC** **40% Use at 80A** **For 10 Continuous Minutes**
2-1/2 Minutes Welding / **7-1/2 Minutes Resting**	**4 Minutes Welding** / **6 Minutes Resting**
100% Continuous Use at 100A	**100% Continuous Use at 60A**

Setting up the Weld

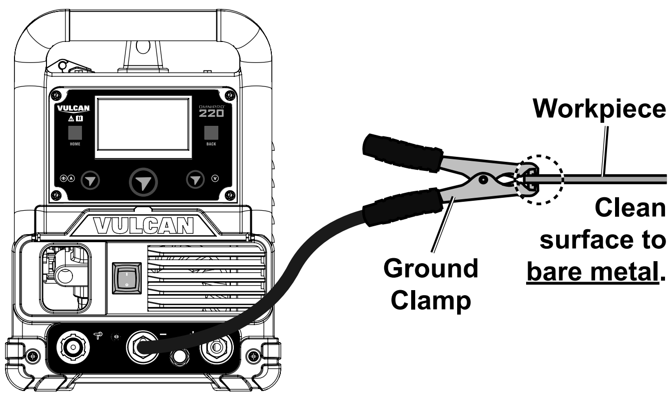


1. **Make practice welds on pieces of scrap the same thickness as your intended workpiece to practice technique before welding anything of value.** Clean the weld surfaces thoroughly with a wire brush or angle grinder; there must be no rust, paint, oil, or other materials on the weld surfaces, only bare metal.

2. Use clamps (not included) to hold the workpieces in position so that you can concentrate on proper welding technique. The distance (if any) between the two workpieces must be controlled properly to allow the weld to hold both sides securely while allowing the weld to penetrate fully into the joint. The edges of thicker workpieces may need to be chamfered (or beveled) to allow proper weld penetration.

NOTICE: When welding equipment on a vehicle, disconnect the vehicle battery power from both the positive connection and the ground before welding. This prevents damage to some vehicle electrical systems and electronics due to the high voltage and high frequency bursts common in welding.

3. Clamp Ground Cable to bare metal on the workpiece near the weld area, or to metal work bench where the workpiece is clamped.



Ground connection depends on desired welding polarity